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DOOR LOCK ASSEMBLY FOR A WASHING MACHINE APPLIANCE

Abstract

A washing machine appliance includes a cabinet including a front panel defining a chamber opening, a wash tub positioned within the cabinet and defining a wash chamber, a wash basket rotatably mounted within the wash tub, a door rotatably mounted to the front panel and moveable between open and closed positions, and a door lock assembly for selectively locking the door in the closed position. The door lock assembly includes a door lock extending within the cabinet and including a latch moveable between locked and unlocked positions and a reset actuator for moving the latch from the locked to the unlocked position, and a reset lever for engaging the reset actuator to move the latch from the locked to the unlocked position. The reset lever is rotatable mounted within the cabinet and the front panel of the cabinet defines an opening for providing access to the reset lever.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to washing machine appliances, or more specifically, to the door lock assembly of a washing machine appliance.

BACKGROUND OF THE INVENTION

[0002] Washing machine appliances generally include a tub for containing water or wash fluid, e.g., water and detergent, bleach, and/or other wash additives. A basket is rotatably mounted within the tub and defines a wash chamber for receipt of articles for washing. Washing machine appliances also generally include a door that can be opened to provide access to the wash chamber and closed to prohibit access to the wash chamber. During normal operation of such washing machine appliances, the wash fluid is directed into the tub and onto articles within the wash chamber of the basket. The basket or an agitation element can rotate at various speeds to agitate articles within the wash chamber, to wring wash fluid from articles within the wash chamber, etc. During a spin or drain cycle, a drain pump assembly may operate to discharge water from within sump.

[0003] Certain conventional washing machine appliances include a door lock assembly for selectively locking or unlocking the door of the washing machine appliance during a wash cycle. Notably, door lock assemblies typically include a manual reset mechanism which allows a user to unlock the door, for example, when there is a loss of power to the washing machine appliance. However, such manual reset mechanisms are located within the washing machine appliance and require disassembly/removal of components of the washing machine appliance to access the manual reset mechanism. Such required disassembly/removal of components to access the manual reset mechanism is complicated and time consuming for users.

[0004] Accordingly, a washing machine appliance having an improved door lock assembly would be desirable. More specifically, a door lock assembly that is capable of being reset without disassembling the washing machine appliance would be particularly beneficial.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

[0006] In one exemplary embodiment, a washing machine appliance is provided including a cabinet comprising a front panel defining a chamber opening, a wash tub positioned within the cabinet and defining a wash chamber, a wash basket rotatably mounted within the wash tub for receiving a load of clothes, a door rotatably mounted to the front panel and moveable between an open position for providing access to the wash chamber and a closed position for prohibiting access to the wash chamber, and a door lock assembly for selectively locking the door in the closed position. The door lock assembly includes a door lock extending within the cabinet. The door lock includes a latch moveable between a locked position and an unlocked position and a reset actuator for moving the latch from the locked position to the unlocked position. The door lock assembly also includes a reset lever for engaging the reset actuator of the door lock to move the latch from the locked position to the unlocked position. The reset lever is rotatably mounted within the cabinet and the front panel of the cabinet defines a reset opening for providing access to the reset lever.

[0007] In another exemplary embodiment, a door lock assembly for a washing machine appliance is provided. The washing machine appliance includes a cabinet comprising a front panel defining a chamber opening, a wash tub positioned within the cabinet and defining a wash chamber, and a door rotatably mounted to the front panel and moveable between an open position for providing access to the wash chamber and a closed position for prohibiting access to the wash chamber. The

door lock assembly includes a door lock extending within the cabinet. The door lock includes a latch moveable between a locked position and an unlocked position and a reset actuator for moving the latch from the locked position to the unlocked position. The door lock assembly also includes a reset lever for engaging the reset actuator of the door lock to move the latch from the locked position to the unlocked position. The reset lever is rotatably mounted within the cabinet and the front panel of the cabinet defines a reset opening for providing access to the reset lever.
[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.
[0010] FIG. **1** provides a perspective view of a washing machine appliance according to an example embodiment of the present subject matter.
[0011] FIG. **2** provides a front view of the example washing machine appliance of FIG. **1** with a door in the open position according to an example embodiment of the present subject matter.
[0012] FIG. **3** provides a side cross-sectional view of the example washing machine appliance of FIG. **1** according to an example embodiment of the present subject matter.
[0013] FIG. **4** provides another perspective view of the example washing machine appliance of FIG. **1** according to an example embodiment of the present subject matter.
[0014] FIG. **5** provides a top, cross-sectional view of an example door lock assembly according to an example embodiment of the present subject matter.
[0015] FIG. **6** provides a front view of the example door lock assembly of FIG. **5** according to an example embodiment of the present subject matter.
[0016] FIG. **7** provides a perspective view of a reset lever of the example door lock assembly of FIG. **5** according to an example embodiment of the present subject matter.
[0017] FIG. **8** provides a rear view of the example door lock assembly of FIG. **5** with a reset actuator of a door lock in an extended position and a reset lever in a disengaged position according to an example embodiment of the present subject matter.
[0018] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.
DETAILED DESCRIPTION OF THE INVENTION
[0019] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.
[0020] As used herein, the terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). Approximating language, as used

herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. For example, the approximating language may refer to being within a 10 percent margin.

[0021] Referring now to the figures, FIG. **1** is a perspective view of an exemplary horizontal axis washing machine appliance **100**, FIG. **2** is a front view of washing machine appliance **100**, FIG. **3** is a side cross-sectional view of washing machine appliance **100**, and FIG. **4** is another perspective view of washing machine appliance **100**. As illustrated, washing machine appliance **100** generally defines a vertical direction V, a lateral direction L, and a transverse direction T, each of which is mutually perpendicular, such that an orthogonal coordinate system is generally defined. Washing machine appliance **100** includes a cabinet **102** that extends between a top **104** and a bottom **106** along the vertical direction V, between a left side **108** and a right side **110** along the lateral direction, and between a front **112** and a rear **114** along the transverse direction T.

[0022] Referring to FIGS. **2** and **3**, a wash basket **120** is rotatably mounted within cabinet **102** such that it is rotatable about an axis of rotation A. A motor **122**, e.g., such as a pancake motor, is in mechanical communication with wash basket **120** to selectively rotate wash basket **120** (e.g., during an agitation or a rinse cycle of washing machine appliance **100**). Wash basket **120** is received within a wash tub **124** and defines a wash chamber **126** that is configured for receipt of articles for washing. The wash tub **124** holds wash and rinse fluids for agitation in wash basket **120** within wash tub **124**. As used herein, “wash fluid” may refer to water, detergent, fabric softener, bleach, or any other suitable wash additive or combination thereof. Indeed, for simplicity of discussion, these terms may all be used interchangeably herein without limiting the present subject matter to any particular “wash fluid.”

[0023] Wash basket **120** may define one or more agitator features that extend into wash chamber **126** to assist in agitation and cleaning articles disposed within wash chamber **126** during operation of washing machine appliance **100**. For example, as illustrated in FIG. **3**, a plurality of ribs **128** extends from basket **120** into wash chamber **126**. In this manner, for example, ribs **128** may lift articles disposed in wash basket **120** during rotation of wash basket **120**.

[0024] Referring generally to FIGS. **1** through **4**, cabinet **102** also includes a front panel **130** which defines a chamber opening **132** that permits user access to wash basket **120** of wash tub **124**. More specifically, washing machine appliance **100** includes a door **134** that is positioned over chamber opening **132** and is rotatably mounted to front panel **130**. In this manner, door **134** permits selective access to chamber opening **132** by being movable between an open position (FIG. **2**) facilitating access to a wash tub **124** and a closed position (FIGS. **1**, **4**) prohibiting access to wash tub **124**.

[0025] A window **136** in door **134** permits viewing of wash basket **120** when door **134** is in the closed position, e.g., during operation of washing machine appliance **100**. As best illustrated in FIG. **4**, door **134** also includes a handle **138** that, e.g., a user may pull when opening and closing door **134**. Further, although door **134** is illustrated as mounted to front panel **130**, it should be appreciated that door **134** may be mounted to another side of cabinet **102** or any other suitable support according to alternative embodiments.

[0026] Referring again to FIG. **3**, wash basket **120** also defines a plurality of perforations **140** in order to facilitate fluid communication between an interior of basket **120** and wash tub **124**. A sump **142** is defined by wash tub **124** at a bottom of wash tub **124** along the vertical direction V. Thus, sump **142** is configured for receipt of and generally collects wash fluid during operation of washing machine appliance **100**. For example, during operation of washing machine appliance **100**, wash fluid may be urged by gravity from basket **120** to sump **142** through plurality of perforations **140**.

[0027] A drain pump assembly **144** is located beneath wash tub **124** and is in fluid communication with sump **142** for periodically discharging soiled wash fluid from washing machine appliance **100**.

Drain pump assembly **144** may generally include a drain pump **146** which is in fluid communication with sump **142** and with an external drain **148** through a drain hose **150**. During a drain cycle, drain pump **146** urges a flow of wash fluid from sump **142**, through drain hose **150**, and to external drain **148**. More specifically, drain pump **146** includes a motor (not shown) which is energized during a drain cycle such that drain pump **146** draws wash fluid from sump **142** and urges it through drain hose **150** to external drain **148**.

[0028] A spout **152** is configured for directing a flow of fluid into wash tub **124**. For example, spout **152** may be in fluid communication with a water supply **154** in order to direct fluid (e.g., clean water or wash fluid) into wash tub **124**. Spout **152** may also be in fluid communication with the sump **142**. For example, pump assembly **144** may direct wash fluid disposed in sump **142** to spout **152** in order to circulate wash fluid in wash tub **124**.

[0029] As illustrated in FIG. **3**, a detergent drawer **156** is slidably mounted within front panel **130**. Detergent drawer **156** receives a wash additive (e.g., detergent, fabric softener, bleach, or any other suitable liquid or powder) and directs the fluid additive to wash tub **124** during operation of washing machine appliance **100**. According to the illustrated embodiment, detergent drawer **156** may also be fluidly coupled to spout **152** to facilitate the complete and accurate dispensing of wash additive.

[0030] In addition, a water supply valve **158** may provide a flow of water from a water supply source (such as a municipal water supply **154**) into detergent drawer **156** and into wash tub **124**. In this manner, water supply valve **158** may generally be operable to supply water into detergent drawer **156** to generate a wash fluid, e.g., for use in a wash cycle, or a flow of fresh water, e.g., for a rinse cycle. It should be appreciated that water supply valve **158** may be positioned at any other suitable location within cabinet **102**. In addition, although water supply valve **158** is described herein as regulating the flow of “wash fluid,” it should be appreciated that this term includes, water, detergent, other additives, or some mixture thereof.

[0031] A control panel **160** including a plurality of input selectors **162** is coupled to front panel **130**. Control panel **160** and input selectors **162** collectively form a user interface input for operator selection of machine cycles and features. For example, in one embodiment, a display **164** indicates selected features, a countdown timer, and/or other items of interest to machine users.

[0032] Operation of washing machine appliance **100** is controlled by a controller or processing device **166** (FIG. **1**) that is operatively coupled to control panel **160** for user manipulation to select washing machine cycles and features. In response to user manipulation of control panel **160**, controller **166** operates the various components of washing machine appliance **100** to execute selected machine cycles and features.

[0033] Controller **166** may include a memory and microprocessor, such as a general or special purpose microprocessor operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. Alternatively, controller **166** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software. Control panel **160** and other components of washing machine appliance **100** may be in communication with controller **166** via one or more signal lines or shared communication busses.

[0034] During operation of washing machine appliance **100**, laundry items are loaded into wash basket **120** through chamber opening **132**, and washing operation is initiated through operator manipulation of input selectors **162**. Wash tub **124** is filled with water, detergent, and/or other fluid additives, e.g., via spout **152** and or detergent drawer **156**. One or more valves (e.g., water supply valve **158**) can be controlled by washing machine appliance **100** to provide for filling wash basket

120 to the appropriate level for the amount of articles being washed and/or rinsed. By way of example for a wash mode, once wash basket **120** is properly filled with fluid, the contents of wash basket **120** can be agitated (e.g., with ribs **128**) for washing of laundry items in wash basket **120**.

[0035] After the agitation phase of the wash cycle is completed, wash tub **124** can be drained. Laundry articles can then be rinsed by again adding fluid to wash tub **124**, depending on the particulars of the cleaning cycle selected by a user. Ribs **128** may again provide agitation within wash basket **120**. One or more spin cycles may also be used. In particular, a spin cycle may be applied after the wash cycle and/or after the rinse cycle in order to wring wash fluid from the articles being washed. During a final spin cycle, basket **120** is rotated at relatively high speeds and drain pump assembly **144** may discharge wash fluid from sump **142**. After articles disposed in wash basket **120** are cleaned, washed, and/or rinsed, the user can remove the articles from wash basket **120**, e.g., by opening door **134** and reaching into wash basket **120** through chamber opening **132**.

[0036] While described in the context of a specific embodiment of horizontal axis washing machine appliance **100**, using the teachings disclosed herein it will be understood that horizontal axis washing machine appliance **100** is provided by way of example only. Other washing machine appliances having different configurations, different appearances, and/or different features may also be utilized with the present subject matter as well, e.g., vertical axis washing machine appliances.

[0037] Referring to FIG. **1**, a schematic diagram of an external communication system **170** will be described according to an exemplary embodiment of the present subject matter. In general, external communication system **170** is configured for permitting interaction, data transfer, and other communications between washing machine appliance **100** and one or more external devices. For example, this communication may be used to provide and receive operating parameters, user instructions or notifications, performance characteristics, user preferences, or any other suitable information for improved performance of washing machine appliance **100**. In addition, it should be appreciated that external communication system **170** may be used to transfer data or other information to improve performance of one or more external devices or appliances and/or improve user interaction with such devices.

[0038] For example, external communication system **170** permits controller **166** of washing machine appliance **100** to communicate with a separate device external to washing machine appliance **100**, referred to generally herein as an external device **172**. As described in more detail below, these communications may be facilitated using a wired or wireless connection, such as via a network **174**. In general, external device **172** may be any suitable device separate from washing machine appliance **100** that is configured to provide and/or receive communications, information, data, or commands from a user. In this regard, external device **172** may be, for example, a personal phone, a smartphone, a tablet, a laptop or personal computer, a wearable device, a smart home system, or another mobile or remote device.

[0039] In addition, a remote server **176** may be in communication with washing machine appliance **100** and/or external device **172** through network **174**. In this regard, for example, remote server **176** may be a cloud-based server **176**, and is thus located at a distant location, such as in a separate state, country, etc. According to an exemplary embodiment, external device **172** may communicate with a remote server **176** over network **174**, such as the Internet, to transmit/receive data or information, provide user inputs, receive user notifications or instructions, interact with or control washing machine appliance **100**, etc. In addition, external device **172** and remote server **176** may communicate with washing machine appliance **100** to communicate similar information.

[0040] In general, communication between washing machine appliance **100**, external device **172**, remote server **176**, and/or other user devices or appliances may be carried using any type of wired or wireless connection and using any suitable type of communication network, non-limiting examples of which are provided below. For example, external device **172** may be in direct or indirect communication with washing machine appliance **100** through any suitable wired or wireless communication connections or interfaces, such as network **174**. For example, network **174** may include one or more of a local area network (LAN), a wide area network (WAN), a personal area network (PAN), the Internet, a cellular network, any other suitable short- or long-range wireless networks, etc. In addition, communications may be transmitted using any suitable communications devices or protocols, such as via Wi-Fi®, Bluetooth®, Zigbee®, wireless radio, laser, infrared, Ethernet type devices and interfaces, etc. In addition, such communication may use a variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

[0041] External communication system **170** is described herein according to an exemplary embodiment of the present subject matter. However, it should be appreciated that the exemplary functions and configurations of external communication system **170** provided herein are used only as examples to facilitate description of aspects of the present subject matter. System configurations may vary, other communication devices may be used to communicate directly or indirectly with one or more associated appliances, other communication protocols and steps may be implemented, etc. These variations and modifications are contemplated as within the scope of the present subject matter.

[0042] Referring now to FIGS. **1** through **2** and **4** through **8**, a door lock assembly **200** that may be used with the washing machine appliance **100** will be described according to example embodiments of the present subject matter. The door lock assembly **200** is used to selectively lock or unlock the door **134** of the washing machine appliance **100** during, for example, a wash or rinse cycle. As such, the door lock assembly **200** includes a door lock **210** (FIGS. **5**, **8**) including a latch **212** (FIGS. **2**, **4**, **6**) moveable between a locked or extended position for locking the door **134** in the closed position, and an unlocked or retracted position for unlocking the door **134**. Notably, the latch **212** of the door lock **210** remains in the closed position while the washing machine appliance **100** is undergoing a wash or rinse cycle and, thus, prevents the door **134** from being opened, even when the washing machine appliance **100** experiences loss of power during the wash or rinse cycle. This prohibits users from opening the door **134** to access the wash chamber **126**.

[0043] In order to permit access to wash chamber **126** in certain situations (e.g., such as a loss of power), the door lock **210** includes a reset mechanism or actuator which may be manually actuated or moved for moving the latch **212** from the locked position to the unlocked position to unlock the door **134** of the washing machine appliance **100**. This allows users to open the door **134** to access the wash chamber **126**. As best illustrated in FIG. **8**, the reset actuator may be a reset pin **214** positioned within an interior **103** of the cabinet **102** behind the front panel **130** and moveable from an extended position to a retracted position to adjust the latch **212** from the locked position to the unlocked position. However, it should be appreciated the reset actuator may be any other suitable actuating component, such as a button, lever, and/or the like. As explained above, since the reset actuator is located within a conventional washing machine appliance, disassembly/removal of components of the washing machine appliance is necessary to access the reset actuator. Such disassembly/removal of components to access the reset actuator is complicated for users. Accordingly, however, aspects of the present subject matter are directed to door lock assembly **200** and features for addressing these and other issues experienced with conventional door lock assemblies.

[0044] According to example embodiments, the door lock **210** of the door lock assembly **200** is positioned and extends within an interior **103** of the cabinet **102** of the washing machine appliance **100**. In some embodiments, the door lock **210** may be mounted to an interior side **105** of the front panel **130** of the cabinet **102**. The front panel **130** may close off or prohibit access to the door lock **210**. However, in alternative embodiments, the door lock **210** may be mounted to a door lock cover **202** via an opening **131** defined by the front panel **130**. The door lock cover **202** may be mounted to an exterior side **107** of the front panel **130** for prohibiting or closing off access to the door lock assembly **200**. Additionally, the door lock cover **202** may be removable for providing access to the door lock assembly **200** when the door **134** is open. Furthermore, when in the locked position, the latch **212** of the door lock **210** may extend through the opening **131** defined by the front panel **130** to engage and lock the door **134**.

[0045] As best illustrated in FIGS. **6** through **8**, the door lock assembly **200** includes a reset lever **220** for engaging the reset pin **214** of the door lock **210** to reset the latch **212** by adjusting the latch **212** from the locked position to the unlocked position. The reset lever **220** may be rotatably mounted to the interior side **105** of the front panel **130** of the cabinet **102** about a pivot joint **204**. The reset lever **220** may thus be rotatable about the pivot joint **204** in a direction of rotation R between a disengaged position (FIG. **8**) and an engaged position (not shown) in which the reset lever **220** moves the reset pin **214** from the extended position to the retracted position. For example, as best illustrated in FIG. **8**, the pivot joint **204** may correspond to a barrel snap **206** protruding from the interior side **105** of the front panel **130**. The barrel snap **206** is received within an opening **221** defined by the reset lever **220**. The reset lever **220** may thus be rotatable about the barrel snap **206** in the direction of rotation R between the disengaged position and the engaged position. However, in embodiments that include the door lock cover **202**, the reset lever **220** may be rotatably mounted to the door lock cover **202**. Additionally, or alternatively, it should be appreciated that the reset lever **220** may move translationally to move the reset pin **214** from the extended position to the retracted position. For example, the reset lever **220** may be slidably mounted to the front panel **130** for moving the reset pin **214** from the extended position to the retracted position.

[0046] According to example embodiments, the reset lever **220** may include a lever body **222** and a lever arm **224** extending or protruding from the lever body **222**. As best illustrated in FIG. **8**, the lever arm **224** may be aligned with the reset pin **214** for engaging the reset pin **214** of the door lock **210**. In this respect, the lever arm **224** applies an increasing load to the reset pin **214** as the reset lever **220** is rotated about the pivot joint **204** for moving the reset pin **214** from the extended position to the retracted position and, thus, adjust the latch **212** from the locked position to the unlocked position.

[0047] As best illustrated in FIG. **7**, the lever body **222** of the reset lever **220** may define a recessed portion or groove **226** for rotating the reset lever **220** between the disengaged position and the engaged position. A groove wall **228** may surround the recessed portion **226**. An object for rotating the reset lever **220**, such as a finger or a tool, may be inserted into the recessed portion **226** for contacting and applying force to the groove wall **228** to rotate the reset lever **220**.

[0048] According to example embodiments, one or more limits stops **230** may limit rotation of the reset lever **220**. As best illustrated in FIG. **8**, a pair of limit stops **230** protrude from the interior side **105** of the front panel **130** of the cabinet **102**. The pair of limit stops **230** limit rotation of the reset lever **220** by preventing gravity from rotating the reset lever **220** beyond the disengaged position. As such, when the reset lever **220** is in the disengaged position, the reset lever **220** rests against the limit stops **230**. This prevents the recessed portion **226** of the lever body **222** from becoming misaligned with, as will be described below, a reset opening or slot **133** for accessing and rotating the reset lever **220** from the exterior of the washing machine appliance **100**.

[0049] According to example embodiments, the reset lever **220** may include a biasing element **240**, such as a leaf spring, for biasing the reset lever **220** in the disengaged position. The biasing element **240** may protrude from the lever body **222** and be detachably coupled to one of the limit stops **230**. For example, the pair of limit stops **230** may be cylindrically shaped and the biasing element **240** may include a hooked portion **242** that attaches around one of the cylindrically shaped limit stop **230** for holding the reset lever **220** in the disengaged position thus preventing the reset lever **220** from moving from the disengaged position. The object that is inserted into the recessed portion of the lever body **222** for moving the reset lever **220** applies force to the groove wall **228** to pull and detach the hooked portion **242** of the biasing element **240** from the limit stop **230** and, thus, rotate the reset lever **220** from the disengaged position.

[0050] According to example embodiments, one or more arced guide rails **250** (FIG. **8**) may protrude from the interior side **105** of the front panel **130** of the cabinet **102** and/or the reset lever **220** for guiding rotation of the reset lever **220** between the disengaged position and the engaged position. The arced guide rails **250** may also limit rotational movement created by movement/vibration during operation of the washing machine appliance **100** for preventing the reset lever **220** from engaging the reset pin **214** of the door lock **210**. Likewise, the arced guide rails **250** may limit translational movement of the reset lever **220**, which may be created by movement/vibration during operation of the washing machine appliance **100**, to prevent misalignment of the lever arm **224** and the reset pin **214** of the door lock **210**. The lever body **222** presses against the arced guide rails **250** that protrude from the interior side **105** of the front panel **130** of the cabinet **102**. Likewise, the arced guide rails **250** that protrude from the lever body **222** press against the interior side **105** of the front panel **130**. The lever body **222** and/or the interior side **105** of the front panel **130** may have corresponding flanges **223** that protrude from the interior side **105** of the front panel **130** and/or the lever body **222** that engage/contact the corresponding arced guide rails **250** as the reset lever **220** is rotated to guide rotation of the reset lever **220**.

[0051] According to example embodiments, the reset opening or slot **133** is defined through the front panel **130** of the cabinet **102** for providing access to the reset lever **220**. As such, a user may access the reset lever **220** from the exterior of the washing machine appliance **100** without disassembly/removal of components of the washing machine appliance **100**. The recessed portion **226** of the lever body **222** may be aligned with the reset opening **133** so that the object for rotating the reset lever **220** may be inserted through the reset opening **133** and into the recessed portion of the reset lever **220**. Additionally, a removable reset opening panel **135** (FIG. **6**) may cover the reset opening **133** for prohibiting/closing off access to the reset opening **133** and, thus, the reset lever **220**. A user may remove the reset opening panel **135** when access to the reset lever **220** is desired.

[0052] According to example embodiments, the door **134** of the washing machine appliance **100** may define a recessed handle portion **137** (FIGS. **4-5**) behind the handle **138** in the direction T. An access gap **139** (FIGS. **4-5**) may be defined between the front panel **130** of the cabinet **102** and the handle **138** of the door **134** for providing access to the reset opening **133** of the front panel **130**. As such, a user may access the reset opening **133** and, thus, the reset lever **220** from the exterior of the washing machine appliance **100** without disassembly/removal of components of the washing machine appliance **100**.

[0053] As explained herein, aspects of the present subject matter are generally directed to a door lock assembly design of a washing machine appliance that includes a door lock and a reset lever accessible via an opening within the front panel of the cabinet of the washing machine appliance. This door lock assembly including the door lock and the reset lever accessible via the opening within the front panel of the cabinet of the washing machine appliance allows users to reset the door lock without removal/disassembly of components (e.g., the front panel, the top panel) of the washing machine appliance to access the manual reset mechanism of the door lock.

[0054] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A washing machine appliance comprising: a cabinet comprising a front panel defining a chamber opening; a wash tub positioned within the cabinet and defining a wash chamber; a wash basket rotatably mounted within the wash tub for receiving a load of clothes; a door rotatably mounted to the front panel and moveable between an open position for providing access to the wash chamber and a closed position for prohibiting access to the wash chamber; and a door lock assembly for selectively locking the door in the closed position, the door lock assembly comprising: a door lock extending within the cabinet, the door lock comprising a latch moveable between an extended position and a retracted position, the door lock also comprising a reset actuator for moving the latch from the locked position to the unlocked position; and a reset lever for engaging the reset actuator of the door lock to move the latch from the locked position to the unlocked position, the reset lever rotatably mounted within the cabinet; and wherein the front panel of the cabinet defines a reset opening for providing access to the reset lever.

2. The washing machine appliance of claim 1, wherein the reset lever is rotatably mounted to the front panel of the cabinet.

3. The washing machine appliance of claim 1, further comprising: a door lock cover mounted to the front panel of the cabinet for prohibiting access to the door lock assembly, and wherein, the reset lever is rotatably mounted to the door lock cover.

4. The washing machine appliance of claim 1, the reset lever including a lever arm for engaging the reset actuator of the door lock to move the latch from the locked position to the unlocked position.

5. The washing machine appliance of claim 4, wherein the reset actuator is moveable from an extended position to a retracted position to move the latch from the locked position to the unlocked position.

6. The washing machine appliance of claim 5, wherein the lever arm of the reset lever is aligned with the reset actuator and the reset lever is rotatable between a disengaged position and an engaged position in which the lever arm moves the reset actuator from the extended position to the retracted position.

7. The washing machine appliance of claim 6, wherein the front panel of the cabinet includes one or more limit stops protruding therefrom for limiting rotation of the reset lever.

8. The washing machine appliance of claim 7, wherein the reset lever includes a biasing element for biasing the reset lever in the disengaged position, the biasing element detachably coupled to one of the limit stops.

9. The washing machine appliance of claim 6, wherein the front panel of the cabinet includes one or more arced guide rails protruding therefrom for guiding rotation of the reset lever between the disengaged position and the engaged position.

10. The washing machine appliance of claim 1, wherein the reset lever defines a recessed portion aligned with the reset opening of the front panel of the cabinet for rotating the reset lever via the reset opening of the front panel.

11. The washing machine appliance of claim 1, wherein the reset actuator is configured as one of a pin, button, or lever moveable from an extended position to a retracted position to move the latch from the locked position to the unlocked position.

12. The washing machine appliance of claim 10, wherein the door defines a recessed portion such that a gap is defined between the front panel of the cabinet and the door for providing access to the reset opening of the front panel.

13. The washing machine appliance of claim 1, wherein: the front panel of the cabinet includes a barrel snap protruding therefrom; an opening is defined within the reset lever for receiving the barrel snap therethrough; and the reset lever is rotatable about the barrel snap.

14. A door lock assembly for a washing machine appliance, the washing machine appliance comprising a cabinet comprising a front panel defining a chamber opening, the washing machine appliance also comprising a wash tub positioned within the cabinet and defining a wash chamber, the washing machine appliance also comprising a door rotatably mounted to the front panel and moveable between an open position for providing access to the wash chamber and a closed position for prohibiting access to the wash chamber, the door lock assembly for selectively locking the door in the closed position, the door lock assembly comprising: a door lock extending within the cabinet, the door lock comprising a latch moveable between a locked position and an unlocked position, the door lock also comprising a reset actuator for moving the latch from the locked position to the unlocked position; and a reset lever for engaging the reset actuator of the door lock to move the latch from the locked position to the unlocked position, the reset lever rotatably mounted within the cabinet; and wherein the front panel of the cabinet defines a reset opening for providing access to the reset lever.

15. The door lock assembly of claim 14, wherein the reset lever is rotatably mounted to the front panel of the cabinet.

16. The door lock assembly of claim 14, the reset lever including a lever arm for engaging the reset actuator of the door lock to move the latch from the locked position to the unlocked position.

17. The door lock assembly of claim 16, wherein the reset actuator is moveable from an extended position to a retracted position to move the latch from the locked position to the unlocked position.

18. The door lock assembly of claim 17, wherein the lever arm of the reset lever is aligned with the reset actuator and the reset lever is rotatable between a disengaged position and an engaged position in which the lever arm moves the reset actuator from the extended position to the retracted position.

19. The door lock assembly of claim 18, wherein the front panel of the cabinet includes one or more limit stops protruding therefrom for limiting rotation of the reset lever.

20. The door lock assembly of claim 14, wherein the reset lever defines a recessed portion aligned with the reset opening of the front panel of the cabinet for rotating the reset lever via the reset opening of the front panel.